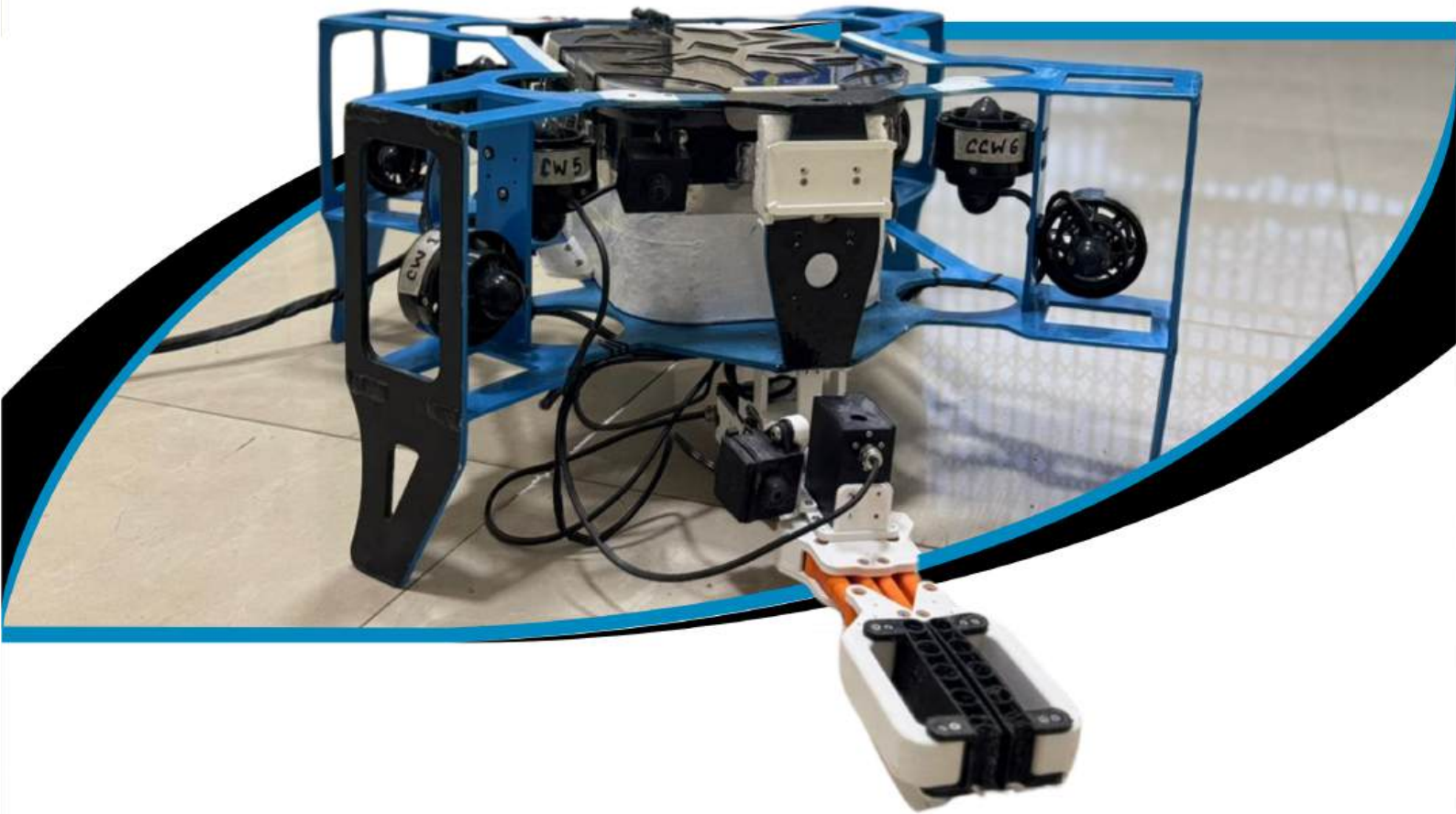




MILITARY INSTITUTE OF SCIENCE AND TECHNOLOGY, DHAKA, BANGLADESH
MATEROV 2026 Technical Documentation



Advisors:

KM Mustafizur Rahman
A N M Didarul Alam
Md Moinul Hasan
M S A Al Farook Shiblee
Md Abdul Wahed
Md Ahsan Kabir
Hasan Sakeef
Md. Soyaeb Hasan

Coach:

Md Naim Uddin

Core Team Members:

Mahibul Haque	(CEO)
Ankon Das Joy	(COO)
Taki Imdadullah Nur	(Design Lead)
Shiekh Tanvir Islam Udo	(Mechanical Lead)
Md. Rokonzaman Shaad	(R & D Lead)
Md. Hasinur Rahman Hasin	(Safety Officer)
Md. Farhan Mohib Hemal	(Software Lead)
Anisul Islam Khan	(Electrical Lead)
Areeb Ayman Haq	(ROV Captain)
Md. Ariful Islam Arif	(CSR Specialist)

Junior Team Members:

Samia Rahman
Nashita Ahmed Chowdhury
Jubayer Ahmed Barno
Tasfia Rahman Priyota

Tasnia Tabassum
Md Jannatul Fardous Emon
Fateen Nur Fuad
Ashik Noor Islam
S.M.Samiul Kabir Saad
Md Rashedul Alam Rafi

Md. Yasin Arafat Labib
Farhan Jobaer Shawpnil
Ishmam Sayeed
Md. Al Amin
A T M Al Nader
Adil Ahnaf Dihan

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পূবালী ব্যাংক পিএলসি
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1. Abstract

MIST Mavirov is a student-led underwater robotics team from the **Military Institute of Science and Technology (MIST)**, Dhaka, Bangladesh. Building on the experience gained from developing two previous ROVs - **Leviathan 2.0**, which refined hydrodynamic stability and system integration and **Creeper 1.0**, which featured wireless control systems, - we present **Mavirio**, our third ROV, developed specifically for the MATE ROV 2026 competition.

Mavirio was designed through three iterative prototypes validated by structural simulation and hydrodynamic drag analysis. The vehicle features a **turtle-inspired** chassis fabricated from **aluminum alloy**, sealed with a custom hybrid gasket formula. An **8-thruster 6-DOF** propulsion system on a **48V DC** architecture delivers full motion control, with compact thrusters providing better thrust-to-power efficiency and finer control resolution over standard T200 units. **Two cameras** have been integrated into the ROV system: a forward-facing camera for forward vision and a downward-facing camera to see beneath the vehicle. Total system weight is around 17 kg with a net positive buoyancy for passive surface recovery.

The electrical system integrates a Pixhawk 4 flight controller, Raspberry Pi 5 and Teensy 4.1 over Ethernet, managed through ground control software. A pressure sensor and a water leak sensor have been used in the system. Software innovations include **hybrid crab species detection pipeline** trained on augmented images, a **lightweight neural network** real-time underwater video enhancement model and **photogrammetry-based 3D mapping**. A 3D-printed 2-DOF gripper with parallel jaw linkage completes the manipulation system.



Fig. 1.1: Leviathan 2.0

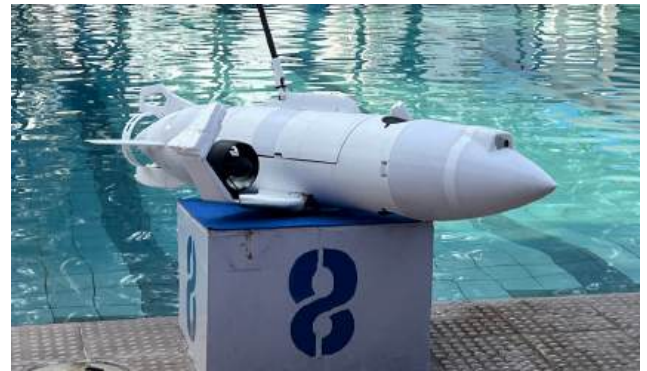


Fig. 1.2: Creeper 1.0



Fig. 1.3: MIST Mavirov Team.

2.Safety

2.1- Company Safety Standards

Safety is fundamental to all ROV design, operation and deployment activities. At **MIST Mavirov**, all team members strictly adhere to established safety protocols to protect personnel, equipment and the environment across all testing, trials and operational activities. Safety features are integrated into all **Mavirov ROVs**.

1. Access and Authorization

- Lab access is restricted to registered **Mavirov** members only.
- Daily check-in/check-out **records** (name, date, operation time) and QR code scanning are mandatory at every entry and exit.
- Visitors require prior approval and must be accompanied by a registered member at all times.
- Members must report to the guard desk or responsible office when required.

2. Lab Conduct and Personal Safety

- All equipment must be operated per **established procedures**; manuals must be consulted where available.
- Food and beverages are **strictly prohibited** near equipment and machinery.
- The lab must remain **clean, organized and hygienic** at all times.
- Appropriate **PPE** (safety glasses, gloves, closed-toe shoes, lab attire) must be worn whenever required.
- Damaged, defective or unsafe tools and components must not be used under any circumstances.

3. Electrical and Equipment Safety

- **Inspect all electrical connections** before operation; avoid exposed wires, loose connections, short circuits and damaged insulation.
- Handle all power sources - batteries, DC sources, and mains -with care; switch off power before any maintenance or repair.
- Dangerous **electrical or mechanical** faults must not be addressed independently.

4. Emergency and Incident Reporting

- Upon any hazard, **immediately halt operations**, cut power and secure the area.
- **Fire extinguishers** and first aid kits must remain accessible at all times.
- All injuries, malfunctions, leakages, unsafe

conditions, or near-miss incidents must be reported without delay.

5. Shutdown and Leaving the Lab

- All systems, equipment and power sources must be **switched off** prior to departure.
- Conduct a **final walkthrough** to confirm no faults, running equipment or unsafe conditions remain.
- **Waste must be disposed** of properly; nothing must be left stored inside the lab.
- Members must report to the guard desk, duty officer, CSM or duty **JSO** when required.



Fig. 2.1: Safety Compliance of Operating Personnel.

2.2- ROV Safety Feature

We have maintained all **MATE**-expected workmanship practices and followed every safety instructions throughout the **ROV** design, fabrication and operation process including

- The aluminium body of the **ROV** has been painted with a protective coating to reduce corrosion and prevent micro metal particles from mixing with seawater.
- A leak **sensor** has been installed inside the watertight housing to detect any water entry and stop operation.
- The high-current internal wires have been **insulated and shielded** to prevent electromagnetic interference.
- **Positive buoyancy** ensures automatic resurfacing of the **ROV** in case of failure or communication loss.
- Software-defined limits **restrict overcurrent**(PWM) to prevent all motor overdrive.

3.Environmental Consideration

Environmental sustainability was one of the primary considerations during the development and manufacturing of the drone. Components and materials were carefully selected after evaluating their potential **environmental impact**. Wherever feasible, **environmentally friendly** and **biodegradable materials** were utilized, including the materials used in the 3D printing process. However, certain specialized components required for underwater operation and system reliability were not available in fully eco-friendly alternatives. Except for these unavoidable components, the majority of the system materials were selected to be **biodegradable and environmentally friendly**.

In addition, a detailed **carbon footprint assessment** of the drone was conducted to evaluate the environmental impact associated with manufacturing, transportation and energy impact (see Table 1). To mitigate these

impact, environmental offset initiatives such as **tree plantation** activities were undertaken by the team as part of its **sustainability commitment**. Further sustainability measures implemented during the project including:-

1. Few of the conventional plastic-based zip ties and binding materials were replaced with biodegradable **jute fiber ropes**.
2. Aquatic plant seedlings commonly found in Bangladesh river ecosystem, such as *Chrysopogon zizanioides*, *Typha latifolia* and *Phragmites australis*, were integrated inside the drone to support biodegradation and assist in absorbing heavy metals and pollutants from river water ,thus promoting environmentally sustainable approach tailored to the river systems of Bangladesh (in case of ROV loss).
3. **Bamboo** was utilized in places as a weight-balancing and **buoyancy-support material** due to its lightweight, renewable and eco-friendly properties.

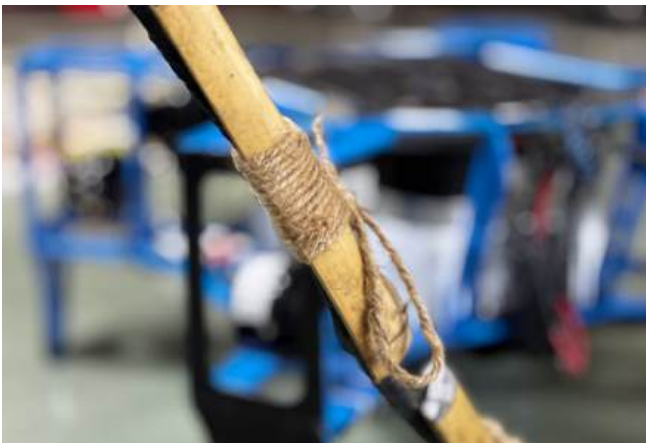


Fig. 3.1: Bamboo Based Tether Buoyancy Control Mechanism.



Fig. 3.2: Aquatic Plant Seedling Integration in Semi-permeable Bag.

ROV Part	Materials /Energy	Quantity	Emission Factor (kg CO ₂ /unit)	Carbon Footprint (kg)
Body	Aluminium 6000 series	10 kg	14.8 / kg [1]	148.0
Wiring	30 m 12 AWG + 26 m 18 AWG (Cu content)	~1.19 kg	4.1 / kg [11]	4.879
3D Printed Parts	PETG filament (3 spools × 1 kg)	3 kg	5.85 / kg [10]	17.6
Energy (build)	Machining, 3D printing	~10 kWh	0.61 / kWh [9]	6.1
Total carbon footprint				176.579
Total Trees Planted		12	-21/kg/year [12]	-252
Net Carbon Emission				-75.421

Table 1: Carbon Footprint Assessment- a Calculation of Carbon Emission and Offset by Tree Plantation.

4. ROV System Architecture and Design Rationale

Our ROV, **Mavirio**, was developed specifically for the **MATE ROV 2026** competition with a strong focus on reliability, modularity and efficient mission execution. The design process was heavily influenced by the team's previous experience from developing earlier ROVs, **Creeper 1.0** and **Leviathan 2.0**. The ROV was initially developed as a **complete 3D model** (see Fig. 4.1), after which simulations including **structural stress-strain analysis** and **hydrodynamic drag** evaluation were conducted. The results obtained from these simulations were used to **iteratively refine and optimize** the design before finalizing the overall system configuration. Before finalizing the current system, **three separate design iterations of the vehicle** were developed and rigorously tested under different operational conditions.

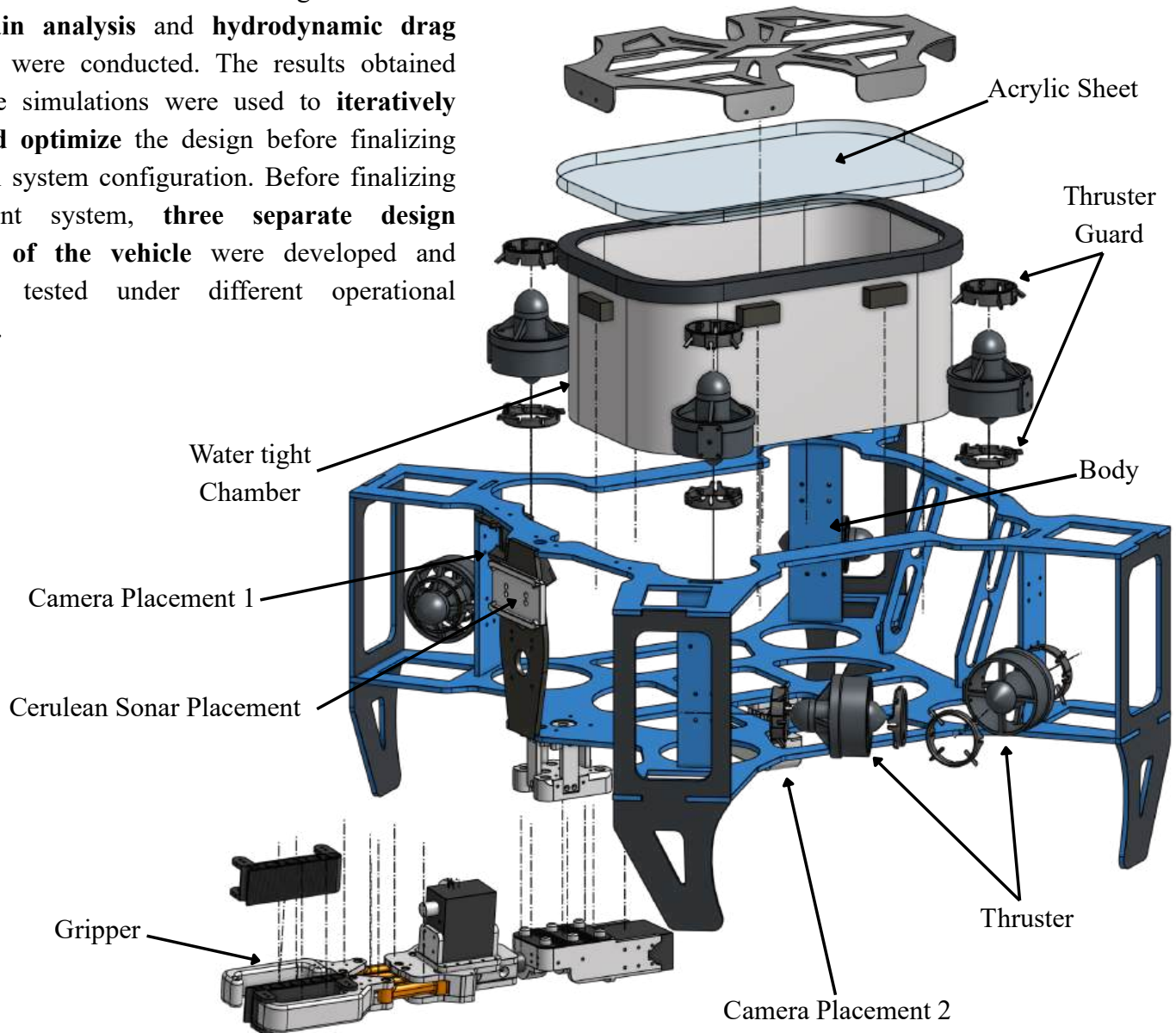


Fig. 4.1: 3D Component View of the ROV System.

Based on the **observations and lessons learned from these tests**, all major components and subsystems of the final ROV were carefully selected. Each subsystem was independently designed in a **modular manner**, allowing rapid replacement or modification in the event of failure during testing and operation. This modular approach also enabled the team to repeatedly **interchange components** between iterations and experimentally evaluate different subsystem combinations. **Multiple refinements and configuration** changes were performed during the final development stage to determine which components worked together most effectively as a complete integrated system. The final ROV configuration is therefore the result of continuous testing, troubleshooting, redesign and practical validation rather than a single-stage theoretical design approach. No **pneumatic-powered** mechanisms or systems were incorporated into the ROV. The following sections contain a detailed discussion of each subsystem.

4.1- Mechanical Subsystem

4.1.1 Outer Chassis

The outer chassis of the ROV is designed using a **biomimetic approach**, inspired by the morphology of a turtle, which naturally exhibits high stability and resistance to overturning. The structure consists of a central protective housing analogous to a turtle shell, surrounded by extended lateral arms that increase the **base footprint** and improve **roll stability**. The chassis is fabricated from **6000 series 5mm aluminum alloy (e.g., 6061-T6)**, selected for its high strength-to-weight ratio, corrosion resistance and weldability. The entire system is **positively buoyant**, needing downward thrust for diving. All joints are **TIG welded** using **ER4043** filler, reducing crack susceptibility and residual stresses to produce a rigid space-frame structure with efficient load distribution. **The extended frame geometry** increases the second moment of area (I), enhancing bending and torsional resistance while minimizing stress concentration.

From a hydrostatic stability perspective, the design ensures that the **center of mass (COM)** is centrally located, while the **center of buoyancy (COB)** is positioned slightly above it due to the elevated central chamber and distributed **buoyant volume**. This satisfies the condition for stable equilibrium, generating a restoring moment that prevents 180° inversion providing passive self-righting capability, analogous to metacentric stability in ships.

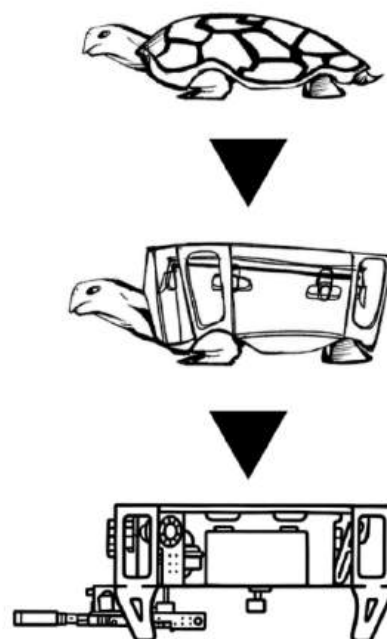


Fig. 4.2: Turtle-Inspired Chassis Design Concept.

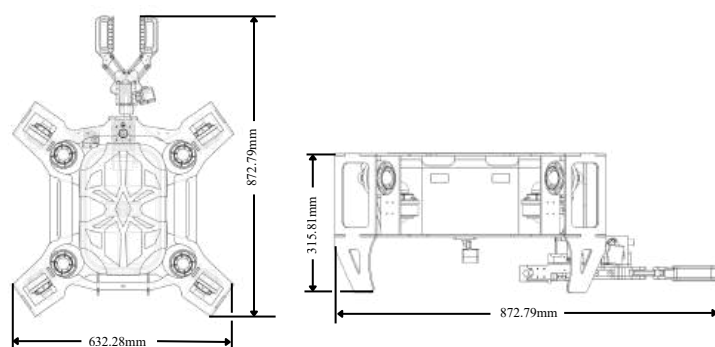


Fig. 4.3: Top and Side Dimensional Views of ROV.

Property	4000 (Al-Si)	5000 (Al-Mg)	6000 (Al-Mg-Si)	7000 (Al-Zn)	8000 (Special alloys)	Mild Steel	Stainless Steel (304)
Yield Strength	8	6	7	10	8	7	6
Strength-to-Weight Ratio	8	8	8	10	9	4	3
Flexibility	7	9	7	5	6	8	7
Cost Efficiency	7	7	9	4	3	10	5
Manufacturing Complexity	5	6	9	5	4	9	6
Thermal Conductivity	7	8	7	6	6	5	3
Corrosion Resistance	6	10	8	5	7	2	10
Total	6.8	7.7	7.8	6.4	6.1	6.4	5.7

Table 2: Comparative Analysis of Material Properties Used in ROV Chassis Design.

****Note:** A weighted relative rating method was used, where materials were comparatively scored from 0–10 based on their performance range for each property.

Hydrodynamically, the semi-open frame reduces the **effective projected area** and added mass, while the curved central body **lowers the drag coefficient**. Moreover, reduced frontal area and streamlined geometry improve efficiency. The chassis supports **modular integration** of thrusters, sensors and electronics; enabling ease of maintenance and optimized load distribution.

4.1.2 Watertight Central Chamber

The central electronics chamber of the ROV is fabricated from a **3 mm thick aluminium 6000-series sheet**, selected for its favorable strength-to-weight ratio, corrosion resistance and thermal conductivity. The quasi-square chamber geometry with rounded edges enhances structural rigidity and pressure distribution under hydrostatic loading (see Fig. 4.4) by reducing stress concentrations while maintaining efficient internal volume. The chamber provides **maximum buoyancy**.

A **hinged mechanism** has been implemented to enable quick opening and closing of the **12 mm tempered acrylic lid** while maintaining secure and uniform sealing. A custom hybrid gasket was developed using a mixture of silicone and rubber to improve sealing durability.

Gasket Composition = 40% Silicone + 60% Rubber

This enhanced formulation provides the following benefits:

- Enhanced **thermal resistance and durability**.
- Improved elastic recovery and **frictional sealing**.
- Increased **durability** under compression cycle.
- Reduced permanent deformation and **aging effects**.

The **friction-based compression sealing mechanism** becomes more effective under increasing hydrostatic pressure, enhancing watertight integrity during deep-water operations.

The **aluminium chamber body** enhances thermal management by efficiently dissipating heat to the surrounding water, **reducing internal temperature** gradients and preventing moisture accumulation inside the enclosure.

Cable routing is implemented through **acrylic interface** instead of the aluminium body to reduce **thermal expansion mismatch**. To prevent water ingress from **cable damage**, the cable insulation is stripped immediately after entry and the conductors are encapsulated with **epoxy resin**, forming an impermeable barrier that prevents water propagation into the electronics compartment.

The chamber is mounted to the main chassis using **M3 fasteners**, providing modularity and ease of maintenance.

Component	Weight(kg)
ROV Aluminium Body	10
Electrical system	5
Gripper	2
Total	17
Buoyancy of ROV Structure	
ROV Chassis Water Displacement Volume	Buoyant Force (in kg)
0.0179 m ³ (Volume, V)	17.9 ($V \times \text{water density}$)
Net Weight in Water	-0.9

Table 3: ROV Weight Analysis and Net Buoyancy Assessment (Archimedes Principle).

****Note:** Actual ROV weight may slightly vary by $\pm 5\%$ due to weight adjustments and use of buoyancy control mechanisms but **will not exceed 18 kg**.

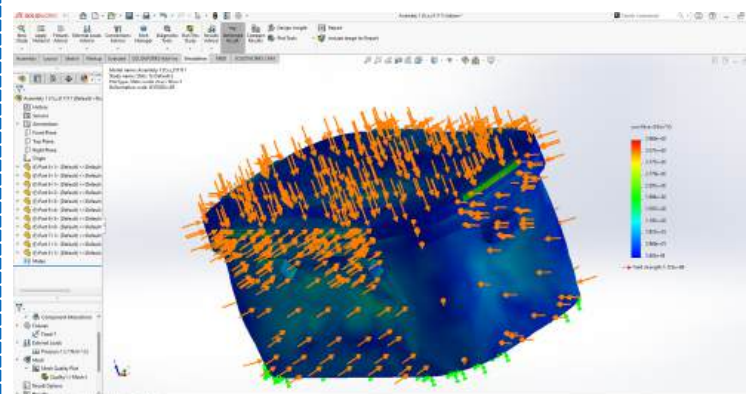


Fig. 4.4: Structural Stress Analysis of Watertight Chamber.

4.1.3 Propulsion System

The chassis has been designed with a **modular mounting** architecture that allows seamless integration with any standard Blue Robotics **T200 Thruster** form factor. This ensures that the system remains future-proof and adaptable, enabling the use of both full-scale T200. In the current configuration, smaller thrusters have been utilized instead of the full-size T200 units. This design choice provides several analytical advantages.

Firstly, smaller thrusters exhibit improved thrust-to-power efficiency under low-load operating conditions. Low-load performance is critical, as the ROV must

operate efficiently within the imposed current limitations. As such, a typical T200 thruster produces approximately 23.5 N thrust at around 144 W (≈ 0.163 N/W), whereas a smaller thruster generating 19.6 N at 72 W achieves approximately 0.272 N/W, indicating an **efficiency improvement** of roughly **66.8%** (see Fig. 4.5).

Secondly, the use of smaller thrusters significantly enhances control resolution. With an **8-bit PWM** control (256 steps), the **thrust increment** per step for a T200 is larger than that of smaller thruster for the same control resolution. This results in finer control, which is critical for precision tasks such as station-keeping and manipulation. Additionally, **structural loading** on the chassis is substantially reduced. A T200 can generate significant **reaction forces**, whereas a smaller thruster produces lower reaction force. This reduces **mechanical stress** on mounting points, improving structural reliability and reducing fatigue on the 6000-series aluminum frame. From an electrical perspective, smaller thrusters operate at significantly lower power and current levels (typically

80-150 W and 5-10 A compared to ~ 350 W and ~ 22 A for T200), allowing the use of lighter wiring, smaller ESCs and reducing thermal load within the watertight enclosure. Furthermore, smaller thrusters generate less hydrodynamic disturbance, minimizing wake turbulence and sediment displacement. This is particularly advantageous for underwater inspection and camera-based tasks, where maintaining clear visibility is essential.

Finally, the reduction in individual thruster weight leads to an overall system weight reduction for an **eight-thruster configuration**. This simplifies **buoyancy management** and improves overall energy efficiency. Thus, while the chassis retains full compatibility with T200-class thrusters, the use of smaller thrusters in the current design, combined with an 8-thruster **6-DOF configuration** (see Fig. 4.7) and protective shrouding, achieves a highly optimized balance between efficiency, controllability, structural integrity and operational precision.

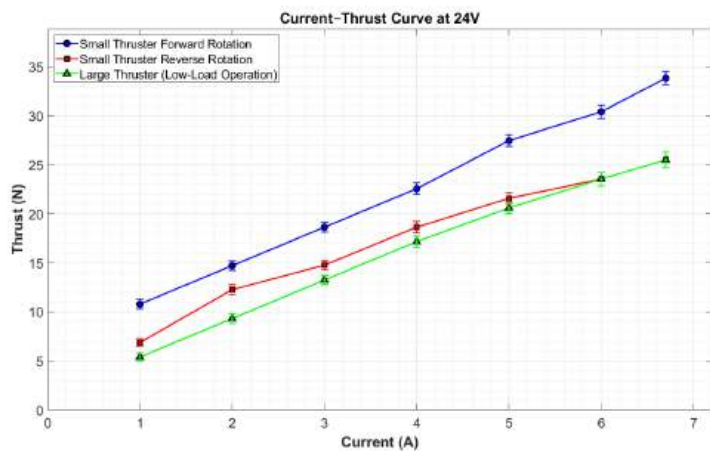


Fig. 4.5: Thrust-Current Performance Curve of ROV Thrusters.

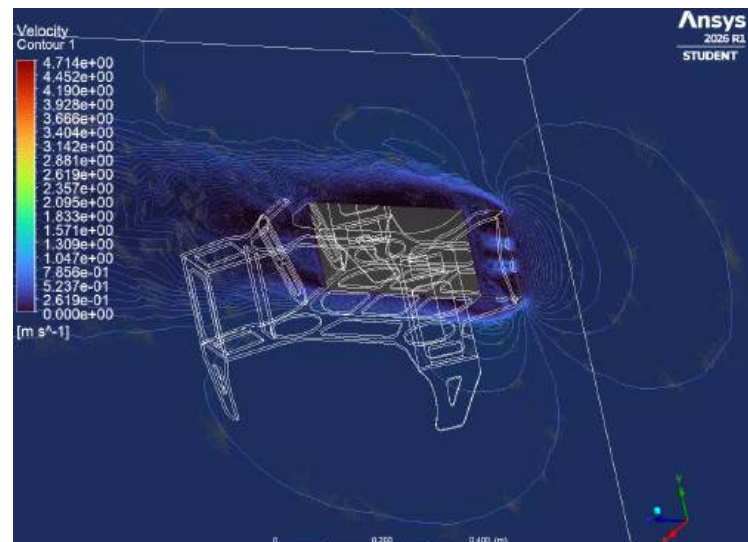


Fig. 4.6: Fluid Velocity Distribution around the ROV Body.

● Up ● Down

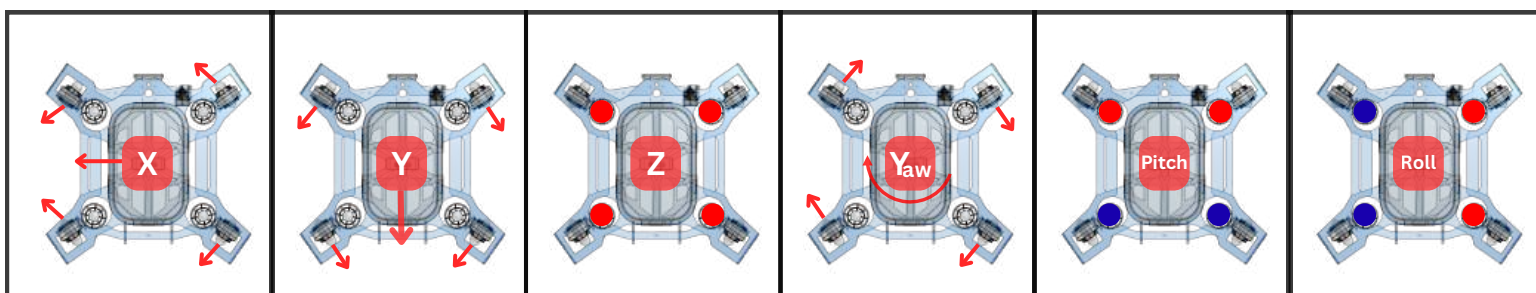


Fig. 4.7: Thruster Force Configuration for 6-DOF (Degrees of Freedom) Motion Control.

4.1.4 Gripper

The ROV gripper is a lightweight, modular and fully **3D-printed 2 DOF** manipulation system designed for reliable underwater operation. The mechanism utilizes a **parallel jaw four-bar linkage** with synchronized spur gear actuation to provide stable and controlled gripping (see Fig. 4.8). A **dual-clamp mounting system** allows quick attachment and removal from the ROV for maintenance and transport. The forward facing wide view camera is used to view the gripper for arm related task.

All major structural components are manufactured using **PETG** due to its low water absorption, good chemical

resistance, strong interlayer adhesion and improved toughness under load. To enhance structural integrity, the printed components are fabricated with **five perimeter wall loops** and **80% infill density**, minimizing internal voids and improving strength for underwater applications.

The overall design follows an **IP20-rated** open mechanical architecture, providing protection against solid object ingress larger than **12.5 mm** while maintaining a lightweight, serviceable and non-sealed structure suitable for controlled underwater robotic operations.

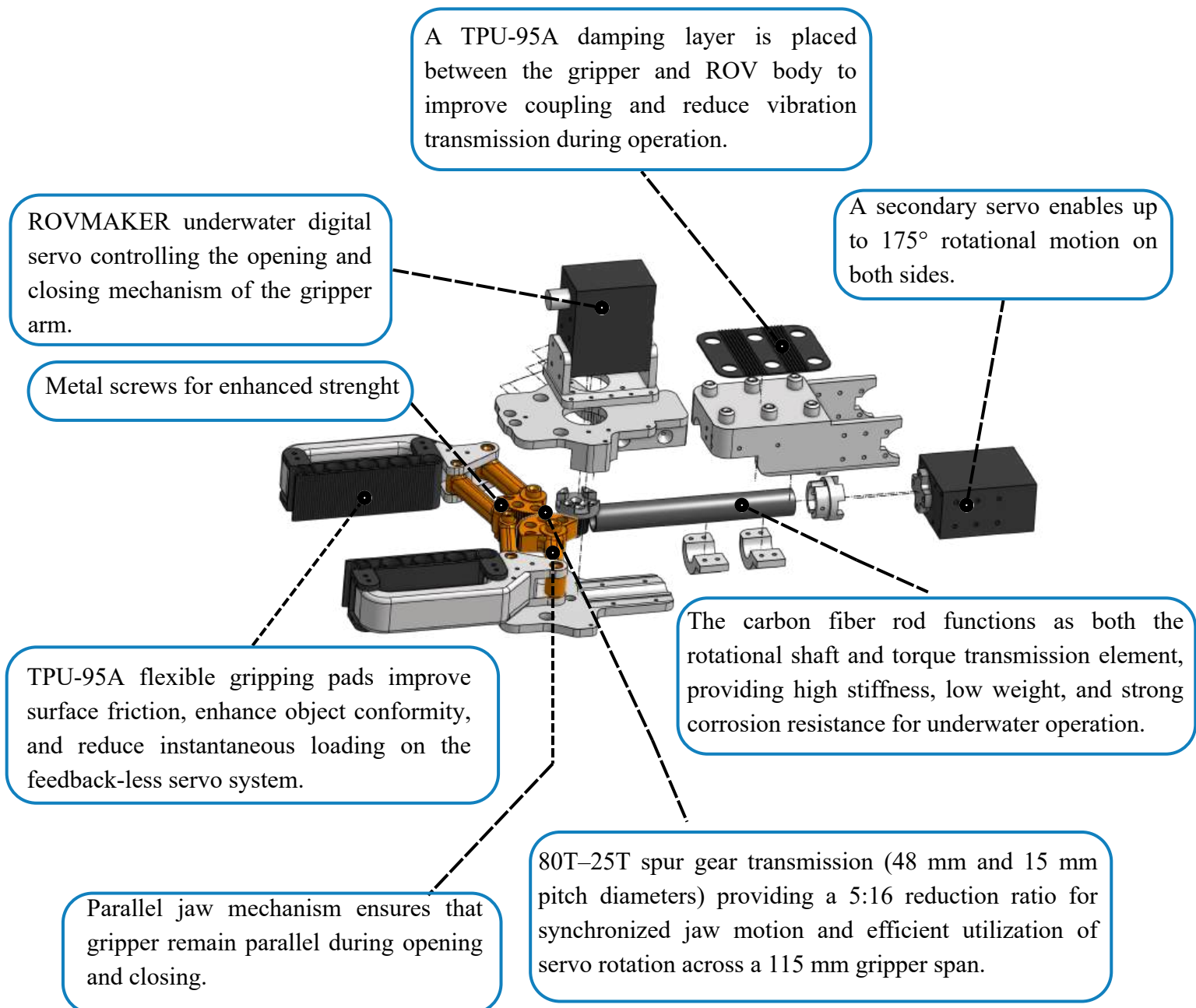


Fig. 4.8: Exploded 3D View of ROV Gripper.

4.2-Electrical Subsystem

4.2.1-System Circuit Architecture

The ROV operates using a **48 V DC** power supply architecture. Two high-power golf cart buck converter modules are employed to step down the 48 V supply to **24 V** and **12 V** rails. The 24 V rail supplies eight 35 A electronic speed controllers (ESCs), which in turn drive eight thrusters. The 12 V rail powers **two servo motors** for the manipulator arm. A secondary buck converter further steps down the **12 V rail to 5 V** to supply the **Teensy 4.1**, which is primarily responsible for arm control. Another dedicated buck steps down 12V for **Raspberry Pi 5 microcontroller**.

Initially, a generic DC-DC buck converter was used to power the **Raspberry Pi 5**; however, this resulted in power throttling and unreliable operation of peripherals such as cameras due to poor voltage regulation, elevated ripple and inadequate transient response under dynamic load conditions. This issue was resolved by employing a **Raspberry Pi PD** buck converter, which incorporates **USB Type-C** compliant power signaling and a high-performance regulation stage. The module ensures a stable **5 V output** with low ripple, high load regulation accuracy and fast transient response, thereby mitigating undervoltage conditions and enabling consistent

processing performance. All cameras are powered directly through the **Raspberry Pi USB interface**, ensuring centralized power distribution for vision subsystems.

For communication, both the Raspberry Pi and the Teensy transmit data serially over **Ethernet** through a network switch, enabling distributed processing and robust data exchange. The Raspberry Pi communicates serially with the **Pixhawk 4**, which is powered via a **PM02 V3 power module** supplied from the 12 V rail. This dedicated power path ensures stable voltage and sufficient current availability for reliable controller operation. The Pixhawk 4 functions as the central control unit, acquiring pressure data via the **I2C protocol**, along with onboard compass and gyroscopic measurements, and generating PWM control signals for all thrusters accordingly.

Additionally, the Teensy 4.1 is interfaced with a **W5500 SPI-to-Ethernet** hardware **TCP/IP** module to enable deterministic and high-speed network communication. The module communicates with the microcontroller via the SPI protocol, offloading TCP/IP stack processing and thereby reducing computational overhead while improving real-time communication reliability within the ROV system.

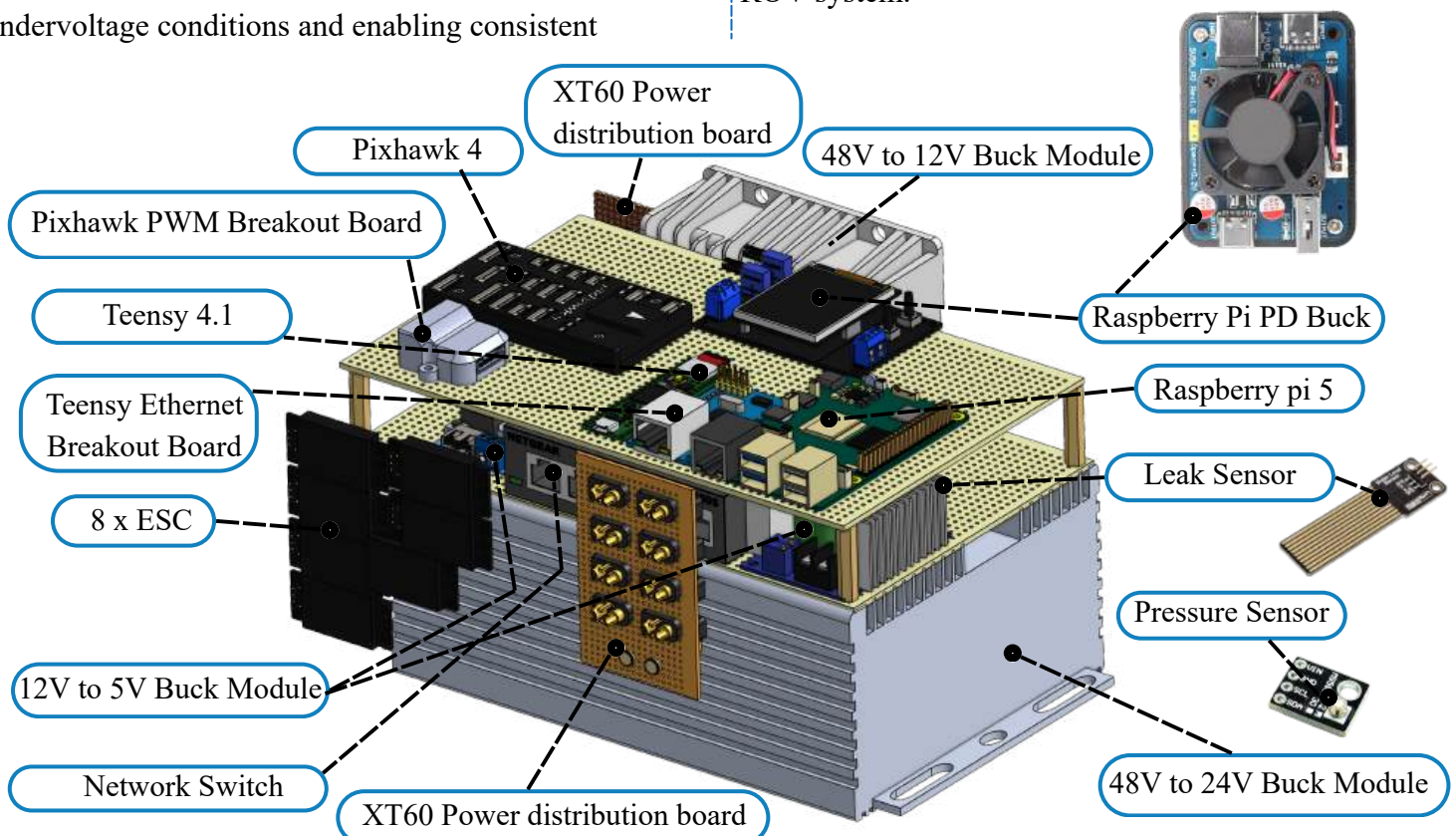


Fig. 4.9: Electronic Compartment Assembly of the ROV Along with the Sensors Used in the System.

4.2.2-Power Distribution

Power enters through an **Anderson Powerpole SBS50** connector and passes through a **25 A Littlefuse JCASE** cartridge fuse sized against a measured full-load current of 21.5 A to guard against overcurrent faults. Custom PETG enclosures ensure **IP-30 ingress protection**, while a **star grounding topology** keeps a clean common ground across all subsystems. Motor power lines are also kept physically **separate** from signal lines to cut down on **electromagnetic noise**.

On the stability side, **bulk capacitors** on both the 24 V and 12 V rails handle the sudden current spikes that thrusters and servomotors tend to produce, and ceramic capacitors take care of high-frequency noise. External cables are wrapped in aluminium foil for **EMI shielding**, then tucked into cable organizers to keep any exposed conductive surfaces safely out of reach.

Condition	Max Ampere	Average Ampere	Active Thrusters
Full Load	21.5A	19A	8
Reverse	14A	12A	4
Forward	15A	12A	4
Lateral	8A	8A	4

Table 4: Electrical Load Analysis of the ROV.

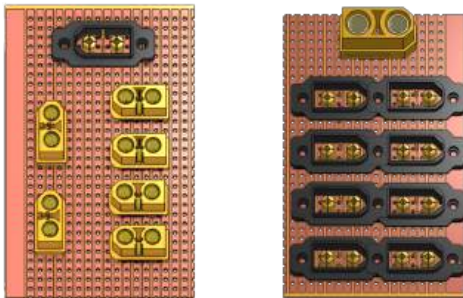


Fig. 4.10: Power Distribution Connector Board.

4.2.3-Sensors

The ROV uses two sensors: one for depth and one for leak detection. Depth measurement and holding are handled by the **GY-MS5837 30BA** pressure sensor, rated for up to 300 meters (see Fig. 4.9). It sits in a custom 3D-printed enclosure outside the main chamber, sealed with epoxy so the circuit stays dry while the sensing element stays in direct contact with the water.

Inside the ROV, a **water level sensor** keeps a constant eye out for any leaks, feeding directly into the Teensy microcontroller as an early warning system.

4.2.4- Tether

The **30-meter tether** pairs a **12 AWG silicone** wire for power with a **CAT6 Ethernet cable** for data, wrapped together in **heat shrink** to keep things tidy and reduce drag (see Fig. 4.11). The topside power connection uses an **Anderson SBS50 connector**.

To keep the tether from fighting the ROV's movements, buoyancy foam & bamboo is distributed along its length, bringing it to **near-neutral buoyancy**. This means the tether floats freely rather than pulling down on the vehicle, hampering its movement.

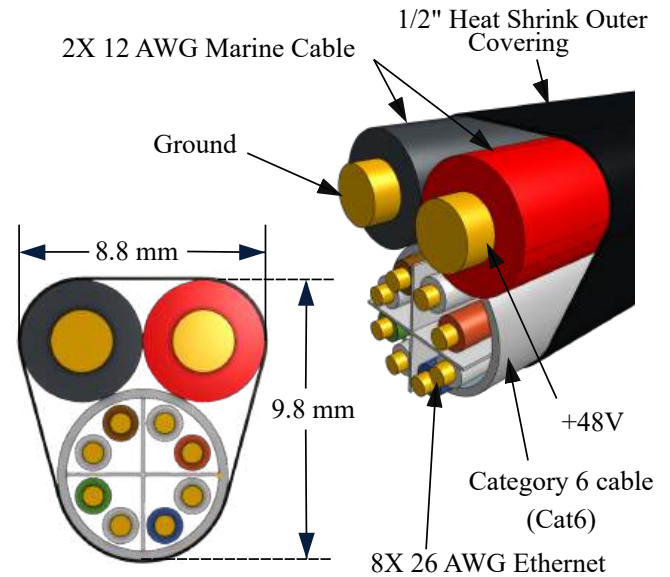


Fig. 4.11: Cross Section of ROV Tether.

4.2.5- Control Station

The control station serves as the surface-side hub for power delivery, communication, monitoring, and mission operation with the ROV. It consists of two AC-powered laptops connected through a **network switch**, where the Ethernet cable from the tether is routed into the switch and then distributed to both laptops. This setup (Fig. 4.12) ensures data connectivity, synchronized communication and efficient handling of different tasks during operation. We use **two laptops** to reduce processing load on a single system and to keep the main **piloting workflow** separate from **mission-specific workloads**. The primary laptop is used for QGroundControl [2], arm control and our custom **ROV GCS software**, allowing the pilot to control the ROV, gripper, monitor telemetry, view system logs and observe camera feeds in real time. The second laptop is used for **auxiliary tasks** such as additional camera monitoring, 3D mapping and computer-vision-based mission activities. By separating these responsibilities,

the pilot laptop remains focused on safe vehicle control, while the second laptop handles heavier processing tasks without affecting the tasks of the main control interface. **PS5 controller** is used to control the ROV (Fig. 4.13).



Fig. 4.12: Topside ROV Control Station.



Fig. 4.13: PS5 Controller Used for ROV Control.

4.2.6- Camera Integration

The ROV's camera system comprises two **USB IMX323** camera - all operating at full **HD 1080p** (1920×1080) under the **H.264** codec for consistent video output. The **IMX323** cameras feature a special wide angle lens suited for underwater inspection. One cameras is front-mounted to provide **150° field of vision** for maneuvering and manipulator control, while a second camera is positioned beneath the ROV for stable observation during fixed-object tasks (Task 1.3).

Each camera is housed in a PETG 3D-printed enclosure fabricated via additive manufacturing (see Fig. 4.14). As PETG exhibits layer-based porosity, a multi-layer sealing approach was implemented: enclosures are **epoxy-filled** for structural rigidity and environmental isolation, followed by a secondary cyanoacrylate application to seal micro-gaps. Since cyanoacrylate can adversely react with curing epoxy, a **neutralizing medium (salt)** was employed to mitigate chemical interactions, ensuring reliable **long-term waterproofing**. To suppress electromagnetic interference, camera wires are twisted to reduce **EMI and signal degradation**, and **ferrite cores** are

incorporated along internal wiring to filter high-frequency noise, ensuring stable and uninterrupted camera operation. The custom **3D-printed enclosures** allow flexible camera positioning and orientation control-an approach informed by experience with the previous **ROV, Leviathan 2.0**, where a fully rigid sealed enclosure for the Logitech camera restricted adjustability and rotation.

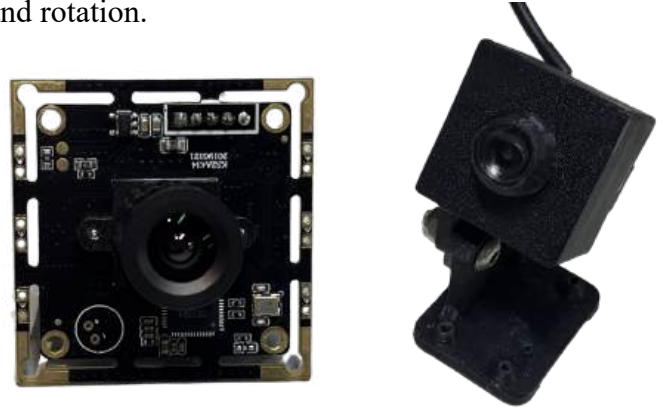


Fig. 4.14: IMX323 ROV Camera and 3D-Printed Watertight Enclosure.

4.3- Software Subsystem

Initially, the widely used **BlueOS platform** was adopted for onboard communication and camera integration. However, to achieve greater flexibility and system control, the architecture was later migrated to **Raspberry Pi OS Image[3]**. In Pi OS Image, multiple USB cameras are streamed through **GStreamer**-based **RTSP** servers, while telemetry from the Pixhawk Flight Controller is forwarded to a ground station running QGround Control using **MAVProxy** over UDP. This approach removed fixed configuration constraints and enabled flexible networking. Direct USB serial communication with Pixhawk provided **lower latency** and stable telemetry transmission, while MAVProxy allowed telemetry to be forwarded simultaneously to multiple ground stations. Consequently, the developed architecture offered a lightweight, flexible and reliable onboard communication system suitable for competition-grade ROV.

4.3.1- System Control Architecture

The ROV system used a distributed architecture consisting of a ground control station, an auxiliary computer and embedded controllers. **Raspberry Pi 5** acted as the **central communication hub** that received telemetry from the **Pixhawk Flight Controller**,

streamed multiple camera feeds and forwarded all data to the pilot computer through a **single Ethernet channel** (see Fig. 4.15). Pixhawk sent detailed telemetry data (depth, sensor information and system state) using the **MAVLink** to the Raspberry Pi via USB. The Raspberry Pi processed this information and forwarded it to the pilot computer using UDP through **MAVProxy**. At the same time, the Raspberry Pi streamed multiple camera feeds using **RTSP**, while the pilot computer sent UDP commands to the **Teensy** microcontroller for **arm control**. All devices were connected through a network switch allowing telemetry, video streams and control commands to travel simultaneously through a single Ethernet cable to the pilot computers. [4]

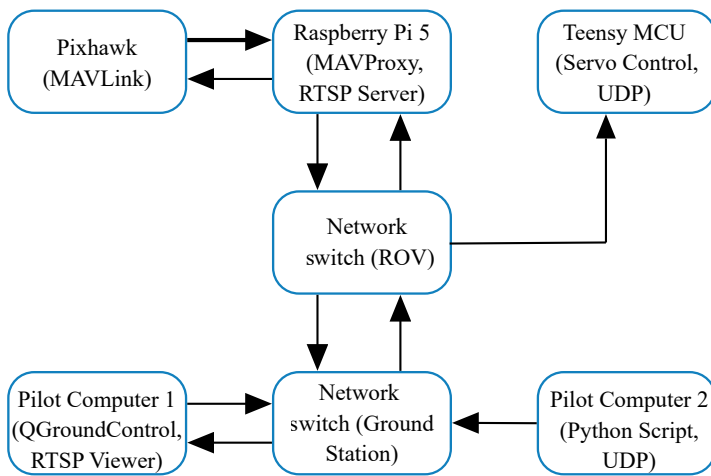


Fig. 4.15: Control Architecture of the ROV.

4.3.2- Arm Control

The Python control script communicates with the Teensy microcontroller over Ethernet using UDP to transmit real-time servo position commands (see Fig. 4.16).

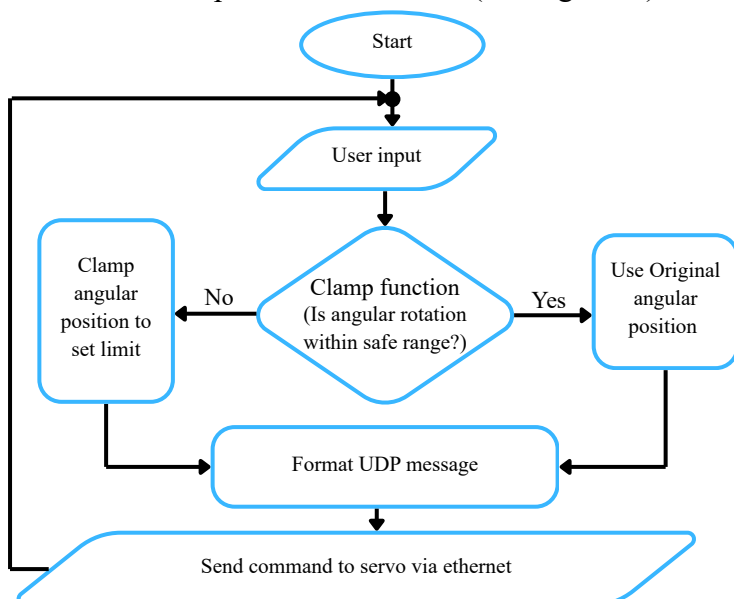


Fig. 4.16: Algorithmic Flowchart for Gripper Control.

4.3.3- Pilot Interface

In the pilot station, we use both **QGroundControl** and our custom-developed software, **ROV GCS**, to make the operation organized. QGroundControl is still used for the main ROV movement because its control response is more precise, stable and dependable for underwater piloting. Alongside it, our ROV GCS works as a dedicated support and monitoring interface where the operator can observe system logs, telemetry data, camera feeds, navigation information, vehicle status and robotic arm operation from one place. The software **displays important flight-controller data** such as roll, pitch, heading, depth, flight mode, arm/disarm condition, leak status, EKF condition, connection messages and other MAVLink-based system information, which helps the pilot understand the real-time condition of the ROV during missions (see Fig. 4.17). The **robotic arm control** has also been integrated into ROV GCS, whereas previously it was operated separately through a Python script in Visual Studio Code. The arm and claw are now controlled through UDP communication with the Teensy-based servo controller, making the pilot station cleaner and easier to manage. For low-latency visual feedback, we use **Happytime RTSP Client.exe**, which receives camera streams from the **Raspberry Pi RTSP** server and provides lower-delay camera viewing compared with our earlier Python-based multi-camera display. This allows the operator to monitor additional camera angles with less delay, while QGroundControl can remain active for the main control view. QGroundControl provides the reliable piloting backbone, while ROV GCS adds our own **customized layer** for monitoring, servo control, system logging and navigation support.

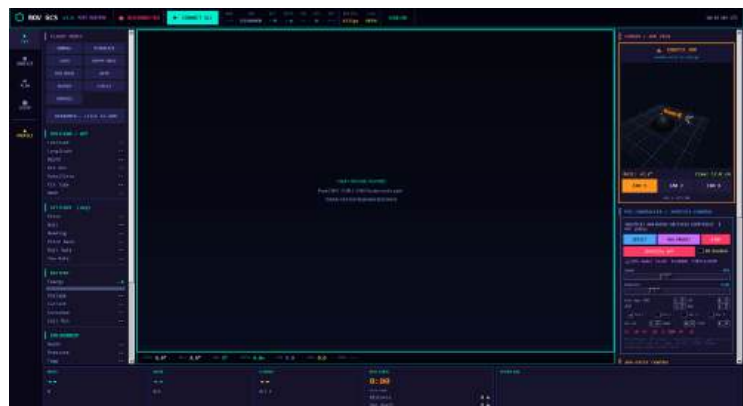


Fig. 4.17: ROV Pilot Monitoring and Control Interface.

4.3.4-Crab Species Detection & Classification

A hybrid computer vision pipeline combining deep learning object detection and colour-space machine learning classification was developed for three-class crab species identification.

Dataset Preparation

As no suitable underwater dataset was available, a **custom dataset** was created by placing **MATE ROV given crab images** underwater and recapturing them. To increase dataset diversity and improve model robustness, an extensive **augmentation** pipeline was implemented using the Python **imgaug** library. Through this process, the dataset was expanded to approximately **16,000 images** for object detection training. The augmentation pipeline included:

- **Geometric transformations:** Flipping, rotation, scaling, cropping and perspective distortion.
- **Image enhancement operations:** Brightness and contrast adjustment, hue and saturation variation, blur and sharpening.
- **Noise and robustness augmentation:** Gaussian noise injection and coarse dropout (random region masking).
- **Spatial deformation techniques:** Affine and elastic transformations.

The pipeline is described below (also see Fig. 4.18):

Stage 1 - Object Detection - RF-DETR Small [5]

RF-DETR Small works as the **primary detection backbone** and trained on the custom augmented dataset for real-time underwater crab detection. The model performed live inference by generating **bounding boxes** around detected crabs under varying underwater conditions including lighting variation, turbidity and complex backgrounds.

Stage 2 - Colour-Based Classification- CMYK Feature Extraction & XGBoost

A hue-based colour-space classification pipeline was developed to **mimic human visual identification** of crab species based on body colour characteristics. The designed classification process is as follows:

- **Bounding Box Crop:** The RF-DETR bounding box was reduced to **50%** of its original size to isolate the crab body and minimise background interference.

- **Colour Feature Extraction:** Mean **RGB** values from the cropped region were extracted and converted to the **CMYK** colour space, where the Cyan and Yellow channels showed **strong species-level discrimination**.
- **Dataset Construction:** A structured **CSV** dataset was created using the extracted CMYK values as input features and crab classes as target labels.
- **Model Training:** An **XGBoost[6]** classifier was trained on the generated dataset due to its fast inference speed, robustness on tabular data and resistance to overfitting for real-time deployment.

Hybrid Inference Pipeline

During live inference, the two models operate in a fused pipeline:

- **RF-DETR Small** performs real-time crab detection and generates bounding boxes with initial class predictions.
- **XGBoost** classifier independently predicts species using **CMYK** color features extracted from the cropped detection region.

The predictions from both models are combined to improve classification reliability by compensating for the domain gap between augmented training data and real-world underwater conditions through the integration of deep learning-based detection and colour-based perceptual reasoning.

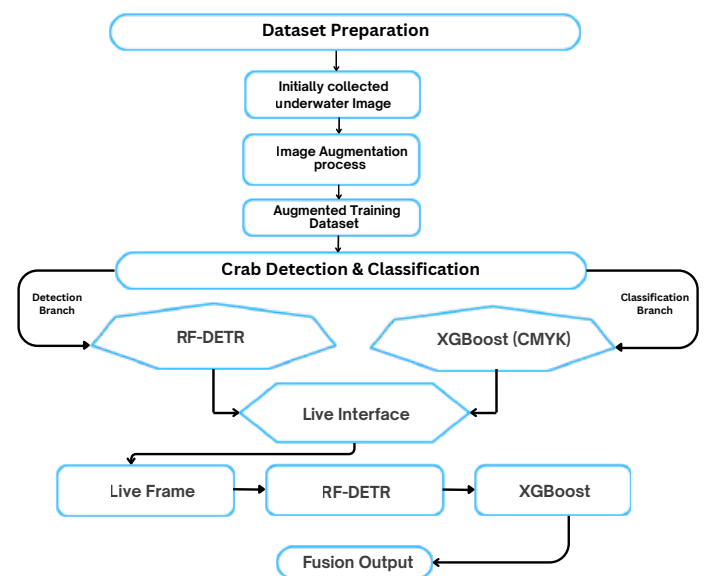


Fig. 4.18: Crab Detection Flowchat.

4.3.5-Real-Time Underwater Video Enhancement

Underwater vision is heavily affected by haze, colour attenuation, scattering and low contrast, degrading both visibility and vision system performance. To address this, a real-time enhancement framework combining DSP preprocessing and a lightweight neural network was developed to properly restore underwater image quality.

Stage 1 - DSP

White Balance Correction is applied to compensate for wavelength-dependent light absorption, which causes dominant blue or green color casts in underwater footage. **CLAHE** (Contrast Limited Adaptive Histogram Equalization) is then used to enhance local contrast and recover surface details in low-visibility regions.

Stage 2 - Lightweight Hybrid CorNet

The core enhancement model is a lightweight CNN designed for real-time inference on constrained onboard hardware. The architecture utilizes **depthwise separable convolutions** to independently perform spatial filtering and channel mixing, significantly reducing **computational complexity** while maintaining enhancement quality. A shallow encoder-decoder structure with residual refinement blocks and attention mechanisms is used to efficiently extract and restore important visual features.

Rather than reconstructing the entire frame, the network operates using **residual learning** by predicting a correction map over the input image. This enables efficient haze suppression, edge and texture recovery, colour consistency refinement, and fine detail enhancement while preserving temporal stability. Together, these operations produce a clearer, natural and visually enhanced underwater image suitable for real-time operation as shown in Fig. 4.19.



Fig. 4.19: Underwater Image Enhancement Results.

4.3.6-Photogrammetry and 3D Mapping

Photogrammetry reconstructs 3D geometry via Structure from Motion (SfM), requiring high image overlap (**>60%**), robust feature detection and accurate camera pose estimation, achieved through **SIFT/SURF** feature matching and bundle adjustment optimization.

Open-source solutions (NeRF, Gaussian Splatting, InstaSplat) **perform poorly** in underwater environments due to rigid algorithmic assumptions, insufficient outlier rejection, poor low-light performance and inadequate distortion handling critical deficiencies given turbidity and light scattering in aquatic conditions.

Softwares Examined	Time	Quality	Noise Tolerance
Custom Library	High	Low	Low
Meshroom	High	Medium	Low
Adobe 3D Sampler	Low	High	Medium
Postshot	Medium	High	High
RealityScan	Low	High	High

Table 5: Comparative Analysis of Different Photogrammetry Software.

Commercial platforms (RealityScan [7], Postshot, Adobe Substance 3D Sampler) **overcome these limitations** through MVS algorithms with advanced preprocessing, incorporating ML-enhanced adaptive feature detection, Levenberg-Marquardt bundle adjustment with automatic outlier filtering, contrast normalization, distortion correction and GPU-accelerated patch-based dense reconstruction. **RealityScan** specifically leverages incremental SfM with hierarchical surface reconstruction for real-time, geometrically accurate output even under sub-optimal imaging conditions (Table 5 and Fig. 4.20).

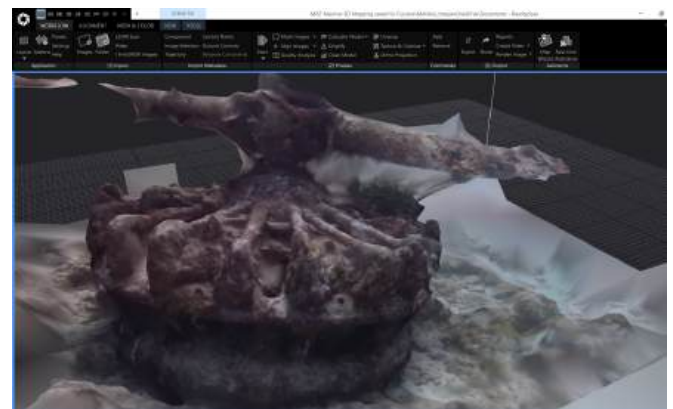


Fig. 4.20: 3D Reconstruction Generated using RealityScan.

5.Floats

The float, **Frost**, was designed with a **latching mechanism** for rapid opening and closing during maintenance, deployment and safety. The system provides secure locking while ensuring uniform sealing pressure to reduce leakage risk and improve servicing efficiency. **No fluid power system or buoyancy tank** has been incorporated into the floats.

All electronic components were mounted on a dedicated removable tray, allowing the complete **electronics assembly** to be installed or removed as a single unit. The tray also acts as a skeleton and provide structural holding point for all electronics compartment.

The platform was designed with **positive buoyancy** so that it naturally returns to the surface in case of power or system failure. This acts as an important passive safety feature during underwater operation. The thrusters were positioned below the watertight chamber to lower the center of gravity relative to the center of buoyancy. This creates a **restoring moment** that naturally keeps the platform upright during operation. Furthermore, heavy components such as batteries and power electronics were positioned strategically to improve balance and underwater stability.

Two thrusters were positioned below the float chamber (see fig. 5) and configured to **rotate in opposite directions** to cancel reaction torque. This reduces unwanted rolling motion and improves operational stability.

A **pressure sensor** was mounted at the top of the float assembly to measure underwater pressure and calculate depth while minimizing disturbance from thruster-generated turbulence using PID control.[8]

The **pressure valve** utilizes a **friction-fit mechanism** in which increasing external underwater pressure further compresses the fitting, resulting in a tighter and more reliable seal during operation. In the event of excessive internal pressure buildup exceeding the retaining force of the fitting, the valve is designed to pop open automatically and release the excess pressure.

The entire system was controlled using an **Arduino Mega** microcontroller, while a 415-915 MHz radio telemetry module transmitted sensor data to the pilot PC after completion of each dive.

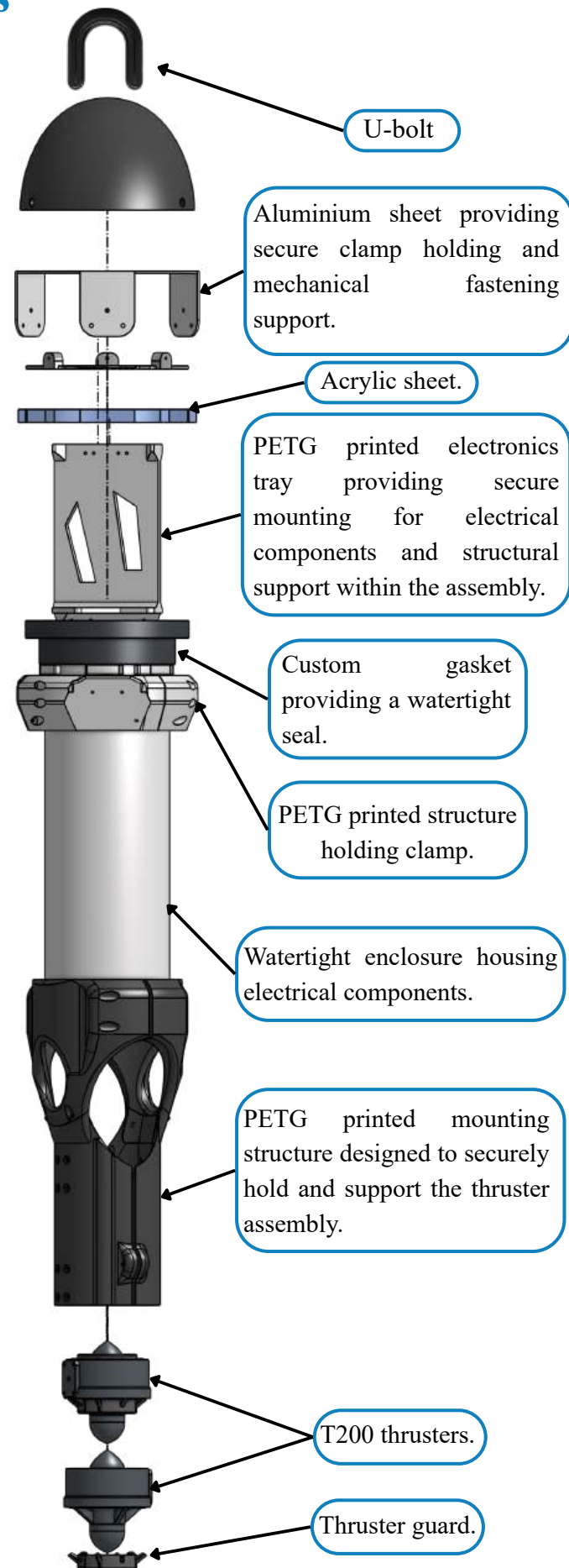


Fig. 5: Exploded View of Floats Assembly.

6. Testing and Troubleshooting

A systematic testing and troubleshooting procedure was developed to rapidly identify, isolate and resolve faults occurring within the ROV system. This approach minimizes operational downtime and ensures reliable performance during missions.

Software Testing and Troubleshooting

- MAVLink **telemetry is monitored** through QGroundControl for heartbeat detection and checking other system parameters.
- The RTSP **camera stream is inspected** for latency, freezing or signal loss.
- Power, thrusters, sensors and cameras are **tested individually** via QGround Control.
- The robotic **arm is independently tested** using a dedicated Python script to evaluate PWM signal reception, movement sensitivity, response accuracy and signal loss.
- In case of **no feedback**, the corresponding component is disconnected and electrically tested independently to verify its operational integrity.

Mechanical Testing and Troubleshooting

- The ROV is submerged to a **depth of 18 ft** and recovered to inspect for internal water ingress, verifying the effectiveness of the watertight sealing system.
- The ROV **orientation is monitored** through QGroundControl in stationary underwater to verify horizontal stability and structural alignment.

- If orientation imbalance is detected, **corrective counterweights are added** to achieve stable and level underwater positioning.
- The robotic arm is operated through its **full rotational range** to verify complete opening and closing functionality of the geared mechanism.
- The flexible TPU section is **mechanically bent** to evaluate whether the required flexibility and deformation characteristics are achieved.
- If any mechanical component **fails to meet operational requirements** during testing, the affected part is reprinted and replaced.

Electrical Testing and Troubleshooting

- A **multimeter** is used to verify that all electrical components receive **the required operating voltage** and to inspect for excessive voltage drops across the system.
- In cases of excessive disturbance, jitter, or unstable operation, the **received signal is analyzed** using an **oscilloscope** to detect electrical noise and signal distortion.
- Components affected by excessive electrical noise are provided with **additional shielding**, filtering circuit and isolation to improve signal integrity.
- Component temperatures are monitored to identify **overheating** or abnormal thermal behavior during operation.
- Electrical circuits are inspected for **water ingress**, short circuits and burn damage caused by leakage or sealing failure.



Fig. 6.1: Experimental Water Testing of the ROV.



Fig. 6.2: Motor Teardown for Troubleshooting.

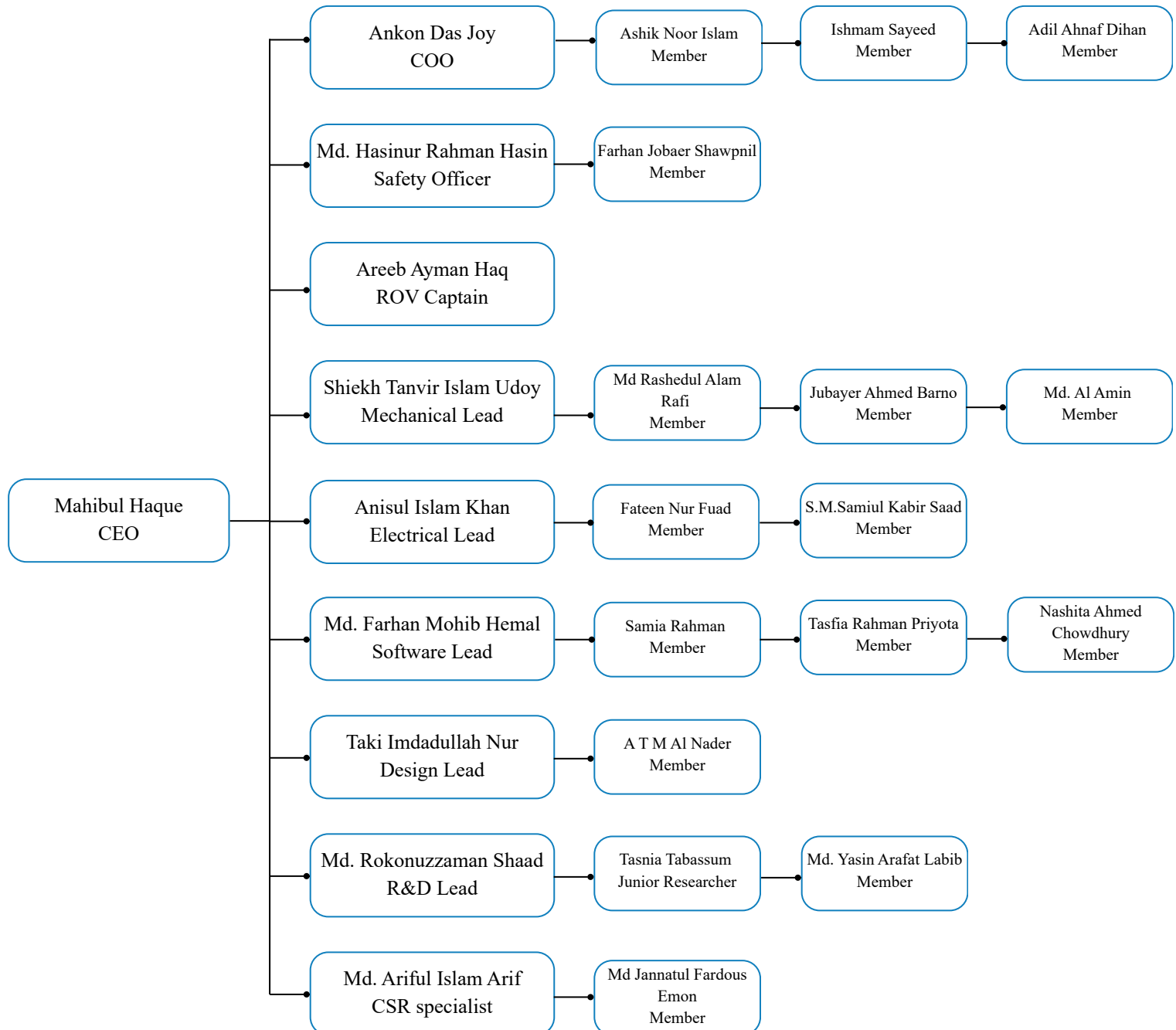
7. Project Organization and Management

7.1- Team Structure

The team follows a **hierarchical** and **function-oriented** structure to ensure efficient management and coordination. The overall project is supervised by a **Chief Executive Officer (CEO)**, who oversees planning, decision-making and interdepartmental coordination.

Each division is managed by a designated **Team Lead** responsible for supervising specific technical or operational tasks. Multiple **junior members** are assigned as General Members under the respective Team Leads, where they contribute to research, design, development, testing and documentation activities within their assigned sectors.

This structure ensures effective communication, clear responsibility distribution and efficient workflow management.



7.2- Budget Overview

The budget analysis reflects the **overall expenditure** throughout our **ROV development journey**. Creeper incurred the highest construction cost due to the integration of **wireless communication** and **autonomous control systems**. Mavirio also required a comparatively higher budget than Leviathan, as several **mission-critical**

components had to be outsourced to ensure reliability and compliance with mission requirements. A detailed budget breakdown and comparative overview are provided on the following page. Previously used lab tools were excluded from the total construction cost of Mavirio.

Sponsorship Donation	
Organization	Amount (USD)
Bangladesh Navy	\$12,000.00
Fresh, Pubali Bank, BOF	\$15,500.00
MIST	\$15,000.00
SKS, Blue Planet Origin	\$5,000.00
City Group, Scube Technologies LTD	\$2,000.00
Total Amount Donated	\$49,500.00
Total Cost (Travel Estimate + Mavirio Cost)	\$40,634.00
Net Remaining (Estimated)	\$8,866.00

Travel Estimate				
Category	Description	Cost (USD)	Qt.	Subtotal
Airfare	Dhaka to St.John's round trip	\$3,000.00	9	\$27,000.00
Transport	Car rental and public transport	\$150.00	9	\$1,350.00
Food	Team food estimate during stay	\$445.00	9	\$4,005.00
Lodging	House / hotel rental	\$550.00	9	\$4,950.00
Total (USD)				\$37,305.00

Cost Comparison Between Earlier and Latest ROV				
Category	Description	Cost (USD) Mavirio	Cost (USD) Creeper-1.0	Cost (USD) Leviathan-2.0
ROV Electronics & Control	Thrusters, Servos, Camera, Converters, Network switch, Connectors, Wires and tether spool; Raspberry Pi, Pixhawk and ESCs.	\$2,348.00	\$1,768.00	\$1,167.00
ROV Mechanical Fabrication	Aluminium frame/cutting, acrylic sheet, epoxy, clamps, stainless cover, carbon fiber rod and paint.	\$315.00	\$1460.00	\$540.00
ROV 3D Printing	Main ROV printed body and support parts.	\$120.00	\$1230.00	\$230.00
Floats Construction Cost				
Float Electronics & Control	Arduino MEGA, Batteries, Converter, Pressure sensor, T200 thrusters, Telemetry, Fuse holder and wire; ESCs.	\$421.00	-	-
Float Mechanical Fabrication	UPVC pipe, Acrylic sheet, Stainless steel, Latch, Gasket and epoxy.	\$75.00	-	-
Float 3D Printing	All printed parts for the float system.	\$50.00	-	-
Total		\$3,329.00	\$4,458.00	\$1,937.00

Built Components			
Category	Component	Cost (USD)	Total (USD)
ROV Body	Aluminium sheet, Screws, Nut	\$210.00	\$929.00
Watertight Chamber	Aluminium sheet, Welding, Acrylic sheet, Silicon glue, Rubber	\$114.00	
Gripper	3D printing filament, Carbon Rod	\$100.00	
Floats Body	UPVC pipe, 3D printing filament, Acrylic sheet	\$125.00	
Tether	18 AWG wire, 12 AWG wire, Ethernet CAT6 cable, Heat Shrink	\$300.00	
Thruster Guard	3D printing filament	\$20.00	
Power Board	XT60 connector, Veroboard	\$60.00	

Reused Components			
Category	Component	Cost (USD)	Total (USD)
Lab Safety Equipments	Safety goggles, Apron, First aid kit, Safety boot, Gloves	\$94.80	\$2,749.60
Lab Equipments	Multimeter, Oscilloscope, Laptop, Monitor, Soldering station, Power supply	\$2,380.00	
Lab Tools	Drill machine, Hand drill, Angle grinder, Hammer, Screwdriver, Plier, Wire Cutter	\$274.80	

Outsourced Components			
Category	Component	Cost (USD)	Justification (Mission Requirement)
Microcontroller	Raspberry Pi5, Teensy 4.1, Pixhawk, Arduino Mega	\$454.00	Computation and ROV control, data streaming, stability, sensor integration.
Motors	T200 thrusters, Servomotor	\$1123.00	Provide thrust and arm control with minimum power, waterproof.
Power Regulators	Buck Modules, Raspberry Pi PD module	\$117.00	Efficient voltage regulation with minimum power loss, good heat dissipation.
Motor Controllers	ESC	\$66.00	Efficient thruster control, can be connected to pixhawk.
Sensor	Pressure sensor, Waterleak sensor, Telemetry Module	\$143.00	High resolution pressure data collection, depth measurement, water ingress detection, floats data streaming to host laptop.
Camera	IMX323	\$322.00	Real time visual feedback and underwater inspection.
Controller	PS5 controller	\$75.00	Good control over ROV motion, pilot familiarity.
Power Connector	Anderson powerpole connector, fuse	\$70.00	Connect ROV power cord to MATEROV DC supply, over current protection.
Control Station	Network switch	\$30.00	Connect multiple microcontroller/laptop together.
Total		\$2,400.00	

7.3 Project Scheduling

At MIST Mavirov, the project workflow begins with defining a clear **central objective** through discussions among team leads. These objectives were further refined with advisors and the team coach to develop a **feasible plan** aligned with available resources and financial constraints. The team strongly believes that clearly defined objectives enhance motivation and guide team performance effectively.

To ensure structured coordination, a dedicated **Discord server** was maintained for communication, alongside **weekly progress meetings**. A centralized **Notion workspace** (Fig. 7.1) was also used to track completed, ongoing and pending tasks, which were initially assigned during meetings involving all team leads.

In the event that any sub-team encounters a **technical or operational challenge**, the respective team **temporarily suspends** related activities until the subsequent weekly coordination meeting. During that session, the issue is collaboratively reviewed and resolved through cross-

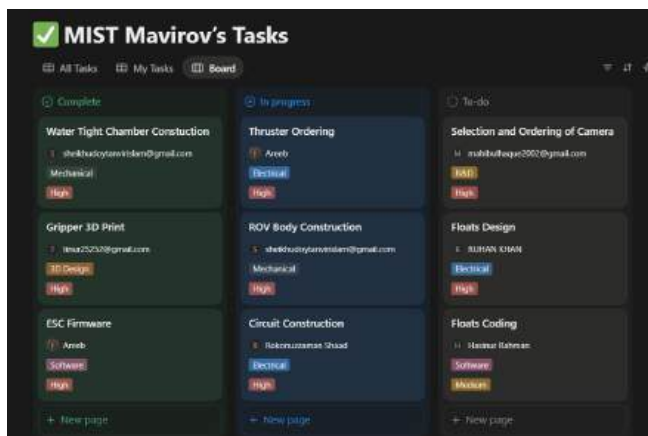


Fig. 7.1: Project Task Tracking Workspace in Notion

team discussion (Fig. 7.2). Following resolution, the updated corrective actions and progress status are updated in the Notion workspace.

During the **procurement** stage, local and international components were identified and sourced while junior members received training under their respective team leads. Simultaneously, the ROV was modeled in **Onshape**, and the design was finalized through collaborative decision-making. Extensive **simulations** were conducted during this phase, and the resulting data were used to optimize the overall design.

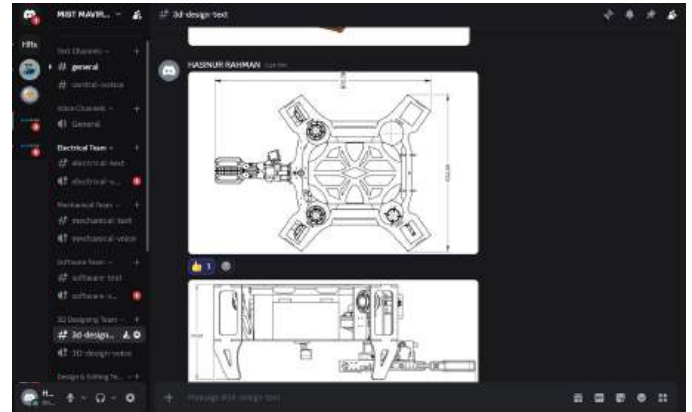


Fig. 7.2: MIST Mavirov Discord Server.

Following design approval, the team divided into **three parallel divisions** focusing on software, electrical and mechanical subsystems. After the release of the MATE ROV competition rulebook in December, assembly and iterative testing commenced. Multiple **prototype iterations** and extensive experimentation enabled the team to identify the most reliable and efficient components for the final ROV configuration. Finally, the completed system underwent comprehensive pool testing to validate operational performance.

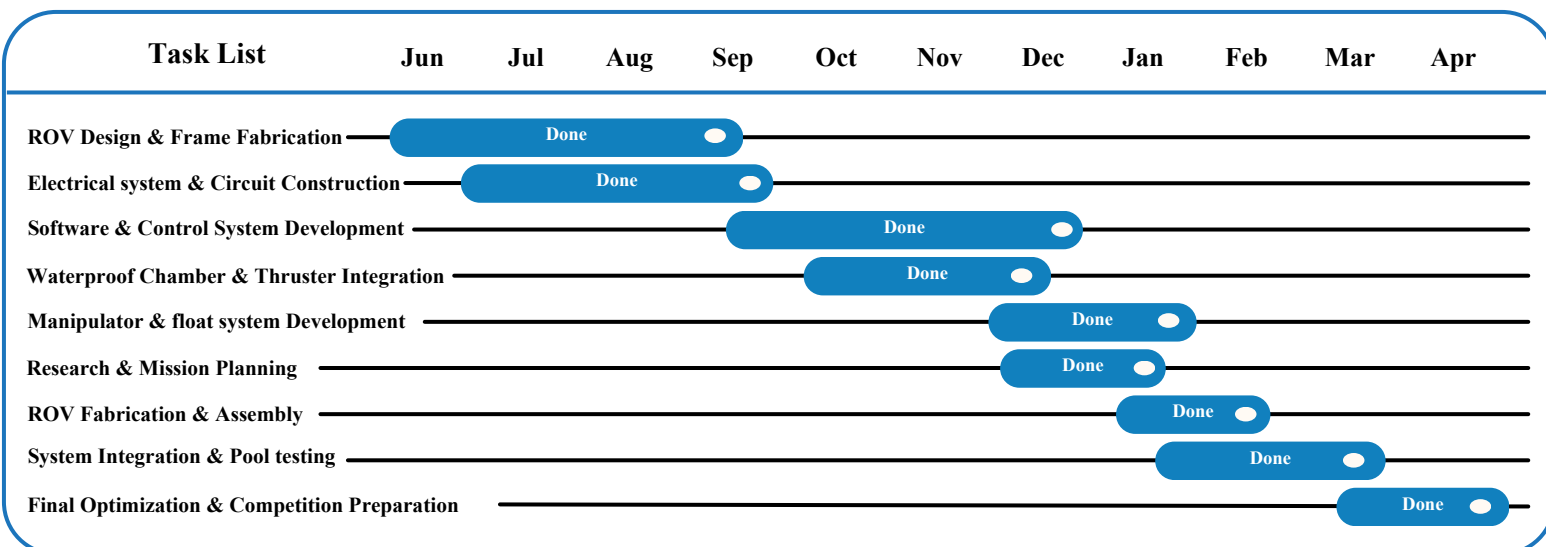


Fig. 7.3: Project Development Timeline of our ROV from June 2025 to April 2026.

8. Key Challenges and Lessons Learned

Import & Resource Limitations

The development of the ROV faced significant financial challenges. Many critical components had to be imported internationally, increasing both procurement time and overall development cost. In addition, the absence of specialized underwater testing facilities forced the team to develop low-cost in-house testing methods capable of producing reliable evaluation results.

Engineering Challenges

- Waterproofing of sensitive electronics.
- Underwater communication reliability.
- Buoyancy and stability control.
- Electromagnetic noise suppression.
- Full system integration under real operating conditions.



Fig. 8.1: Watertight Chamber Prototype.

Among these, waterproofing became one of the most critical and stressful tasks, as even a minor leakage could completely damage all onboard electrical systems. As a result, the waterproof chamber became one of the most rigorously tested components of the entire ROV.

Key Lesson Learned: Underwater chambers should preferably use a single continuous body or sheet structure, as excessive welding distorts and weakens the structure and increases the risk of water ingress.

Manipulator Arm Development

The manipulator arm underwent multiple redesigns to ensure safe interaction with objects of varying shapes and materials without damaging either the object or the environment.

Key Lesson Learned: Manipulator gripping components should have some degree of flexibility. Overly rigid mechanisms directly transfer reaction forces to servo gears, significantly increasing the risk of servo and gear failure.



Fig. 8.2: Iterative Development of the Manipulator Arm Design.

Electromagnetic Noise & Signal Stability

Another major challenge was electromagnetic interference inside sealed underwater compartments. Since enclosed chambers naturally trap and circulate electrical noise, unexpected signal instability issues frequently occurred during operation.

To overcome this, the team implemented:

- Specialized shielding materials.
- Proper grounding techniques.
- Ferrite-core-based suppression systems.

These solutions were specifically adapted for underwater robotic environments to improve communication stability and system reliability.



Fig. 8.3: Cable Noise Suppression using Ferrite Cores.

Growth, Experience and Reflection

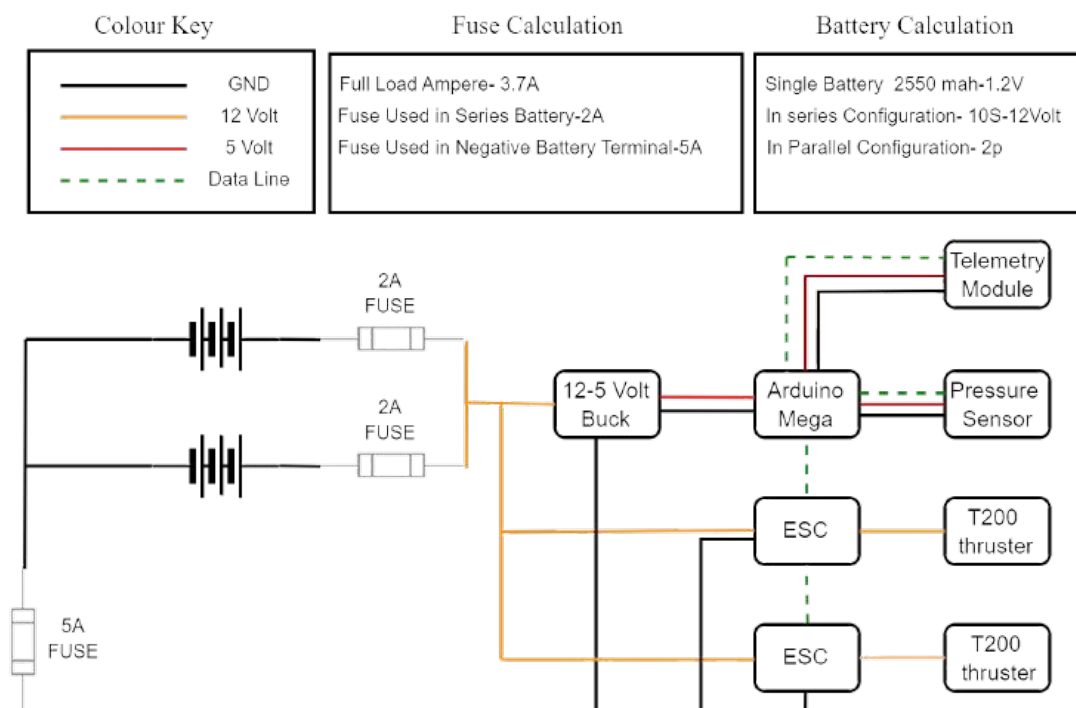
Despite repeated failures, redesigns and underwater trials, the journey transformed MIST Mavirov into a disciplined and research-driven engineering team. The project provided valuable hands-on experience in robotics, embedded systems, fabrication and real-world underwater operations while teaching every member the importance of persistence, teamwork and problem-solving under pressure. Ultimately, the ROV became more than just a machine - it became a symbol of resilience, growth and determination to push beyond limitations.

9. Reference

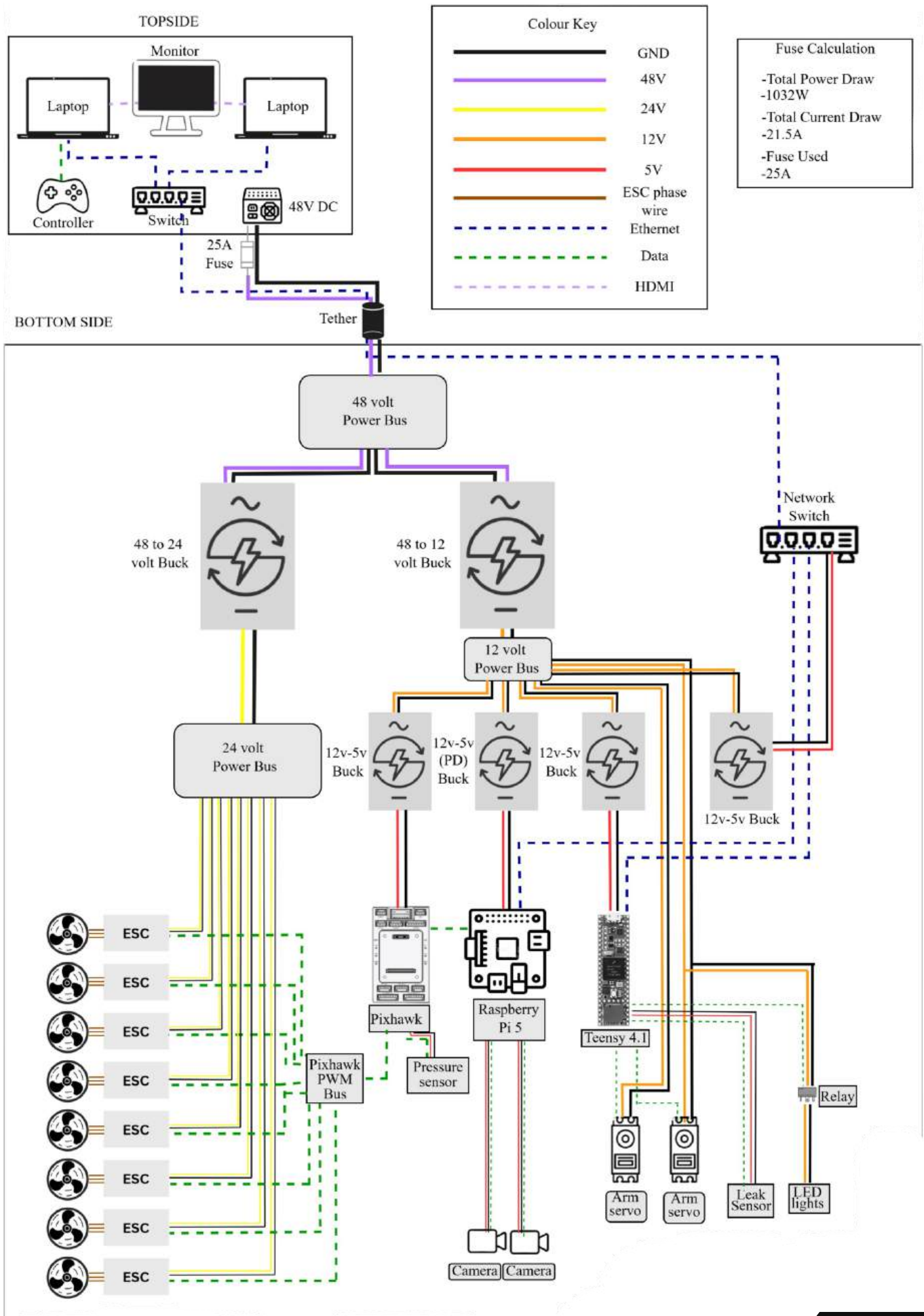
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10. Appendix

10.1- Floats Electrical SID



10.2- ROV Electrical SID



10.3- Safety Checklist

Construction / Assembly Check	
☐ Workbench and ROV exterior are dry before opening/closing.	☐ Route wires neatly so they cannot be pinched by lid
☐ Clear debris, hair, dust and metal particles from the work area.	☐ Clean O-rings and sealing grooves.
☐ Inspect frame for cracks, bent or sharp edges.	☐ Ensure no dirt, hair or metal shaving is on the O-ring surface.
☐ Check thruster guards/shrouds are tight and undamaged.	☐ Align lid evenly with the pressure hull before locking.
☐ Confirm buoyancy.	☐ Lock all lid latches fully and confirm even compression.
☐ Check watertight chamber for crack or deep scratches.	☐ Visually confirm complete O-ring seal around the housing.
☐ Keep inside of watertight chamber completely dry before sealing.	☐ Check internal assembly is level and components are not loose.
☐ Verify internal connectors are fully seated and tight.	☐ Keep tools and loose materials away from electrical hardware.
☐ Confirm no exposed conductor or damaged insulation is visible.	☐ Use gloves/safety glasses during cutting, drilling or grinding.
Pre-Deployment Check	
☐ Poolside area is clear of clutter, loose tools and trip hazards.	☐ Adjust camera position for better visibility if needed
☐ Power supply is dry and kept away from water edge.	☐ Manipulator arm is properly locked in before entering water.
☐ Surface laptop/control station is powered on and connected.	☐ Compass calibration is OK before launch.
☐ Confirm cameras, arm and mission tools are externally secure.	☐ Verify thruster direction with controller input.
☐ Poolside dry run: test every thruster before deployment.	☐ Leak sensor is active and no water is inside the pressure hull.
☐ All pressure-housing lid latches are locked properly.	☐ Pixhawk LED indicator shows the expected status.
☐ Pixhawk gives the correct startup tone.	☐ Team uses closed-toe shoes, safety glasses and tied-back hair.
☐ All camera feeds are visible on the control station.	☐ Confirm communication words before hand-off and launch.
In-water Operation	
Decksider operation	Tether Management Protocol
☐ Pilot announces system readiness before water entry.	☐ Designate someone as tether tender for the duration of operations.
☐ Handler lowers ROV slowly into the deployment zone.	☐ Tether tender uncoils tether in a figure eight on the deck, preventing tangling.
☐ Keep hands clear of thrusters during initialization.	☐ Tether is connected to the surface station strain relief and power and ethernet.
☐ Hold ROV until camera view and control response are stable.	☐ Check strain relief on both ROV and surface side.
☐ Release only after pilot confirms READY FOR HANDS OFF.	☐ Provide enough tether length for the ROV to reach its working depth.
☐ Pilot uses smooth movements; avoid sudden aggressive thrust.	☐ Avoid excess slack; too little inhibits the ROV, too much causes tangling.
☐ Operation leads maintain communication between stations.	☐ ROV pilot avoids 360 degree rotations and close maneuvers around obstacles.
☐ Monitor leak sensor, camera feed and response continuously.	☐ Never step on the tether; it may damage signal and power wires.
☐ If bubbles increase, pull ROV to the surface for inspection.	☐ After operation, tether manager disconnects tether from surface station and power.
☐ If video/control drops, stop motion and cycle power only if safe.	☐ After disconnection, the tether manager coils the tether neatly.
☐ If communication is not restored, power down and recover ROV.	☐ Keep non-essential people away from tether staging zone.
Recovery / Maintenance	
☐ Pilot returns ROV to launch/recovery zone before lifting.	☐ Check camera mounts and adjust if shifted during operation.
☐ Pilot announces READY FOR HANDS ON before recovery.	☐ Inspect arm/tool lock and payload attachment points.
☐ Handler grips frame securely and announces HANDS ON.	☐ Rinse ROV with fresh water after saltwater operation.
☐ Power down before inspecting electrical components.	☐ Wash thrusters and exposed metal parts with clean water.
☐ Check watertight chamber for water, fogging or condensation.	☐ Dry ROV and surface station equipment before storage.
☐ Inspect the inside of watertight chamber removing ROV from water.	☐ Report and repair all damage before the next deployment.
☐ Check lid latches, vent plug and O-ring seal after run.	☐ Store power supply separately in a dry location.
☐ Look for damaged insulation, bent pins or loose connectors.	☐ Save notes from the run for maintenance tracking.
☐ Verify thrusters are free of foreign objects and weed/string.	☐ Return tools and recovery equipment to storage area.